TR No. ELIE-ISE-TD 13-033-1

DEPARTMENT OF THE ARMY

UNITED STATES ARMY INFORMATION SYSTEMS ENGINEERING COMMAND
FORT HUACHUCA, ARIZONA 85613-5300

INSTRUCTIONAL GUIDE FOR DEVELOPING ENGINEERING IMPLEMENTATION PLAN

MAY 2013

U.S. ARMY INFORMATION SYSTEMS ENGINEERING COMMAND

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Technical Director
U.S. Army Information Systems Engineering Command

EXECUTIVE SUMMARY

This document establishes uniform practices for the preparation of Engineering Implementation Plan (EIPs) within the U.S. Army Information Systems Engineering Command (USAISEC).

This guide replaces TR No. ELIE-ISE-TD 13-033, *Instructional Guide for Developing Engineering Installation Packages,* March 2013. This revision was made to change the EIP definition.

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INSTRUCTIONAL GUIDE FOR DEVELOPING ENGINEERING IMPLEMENTATION PLAN

1.0 INTRODUCTION

1.1 Purpose

This document establishes uniform practices for the preparation of Engineering Implementation Plan (EIPs) within U.S. Army Information Systems Engineering Command (USAISEC). An EIP is the technical documentation that translates a validated requirement into an engineered plan. The plan contains the necessary programmatic details and all the detailed installation steps, engineering drawings, Lists of Materials (LOMs), and testing criteria. The EIP normally consists of four major sections, described as follows:

a. Project Overview: This section provides a brief description of the project for the installing activity's use to determine the required capabilities, team size, and timeframe.

b. Implementation Team Responsibilities: This section identifies actions and responsibilities the implementation team must perform for the project. Typical actions include the following:

- Review of site preparation that was included in the Project Concurrence Memorandum (PCM)
- Initial inventory and inspection of the LOM
- Final inventory of excess materials
- Redlining of drawings
- Approval process for changes
- Directions for the use of spares to complete the installation
- Verification, Quality Control (QC), and test responsibilities
- Points of Contact (POCs) for problems with materials
- After action reports

c. The POCs: This section lists the POC information the implementation team may need for engineering, material developers, project management, and others as required.

d. Appendices: This section identifies and describes the appendices included for the project. The following lists the typical appendices:

- Installation Instructions
- Installation Drawings
- LOM
- Validation/Test Information

Note: An EIP is written for the implementation team and Onsite Engineer (OSE). An EIP should provide enough detail to ensure success of the project.

1.2 Scope

The scope of this document is to assist USAISEC personnel in the development of an EIP.

1.3 References

When using an EIP on a contract task order, do not include references listed in the base contract. Use the following references in the development of an EIP, and include applicable references in the EIP (e.g., the PCM or System Design Plan [SDP]):

a. USAISEC, *Standing Operating Procedure (SOP) for Engineering Drawings*, Draft.

b. USAISEC, *Installation Workmanship Standards*, Draft.

c. USAISEC, Technical Director, *Letter of Instruction for Standardized USAISEC Technical Report and White Paper Style Guide*, November 2011.

2.0 DEVELOPMENT OF THE EIP

This section describes the basic document format and the various appendices used in the development of an EIP. An EIP should include a cover page, signature page, and disclaimer page. Use the most recent USAISEC format available. Refer to *Letter of Instruction for Standardized USAISEC Technical Report and White Paper Style Guide* for further information.

2.1 Project Overview

When developing this section, use information from the PCM, Sections 1.0 Programmatic Direction, 1.1 Scope, and 1.2 Site Impact. The project overview provides a brief description of the project for the installing activity to be able to determine the required skills, team size, and timeframe. Example 1 shows a project overview.

Example 1. Project Overview

> **1.0 Project Overview**
>
> To extend Defense Information Systems Network (DISN) services from the Arifjan Technical Control Facility (TCF) to the Earth Terminal Complex (ETC), a fiber optic Interconnect Facility (ICF) will be implemented. Fiber optic modems will be located in the ETC and TCF and will be interconnected using single-mode and multimode fiber. See Appendices B and C for a description of the ICF configuration and the components to be installed at Arifjan.

2.2 Implementation Team Responsibilities

This section identifies the detailed actions and responsibilities the implementation team must perform during the project. The responsibilities should include tasks, such as performing a LOM inventory and an inspection to identify any items that have not arrived or are damaged, as well as producing redline drawings. The section should also cover the processes for approving changes to the installation, using spares to complete the installation, identifying POCs for problems with material-developer-provided resources, and producing after action reports to identify deficiencies or problems.

It is the implementation team's responsibility to install all equipment to USAISEC standards. An EIP should cover safety items in this section, along with any special instructions to OSEs. This section should include details for storing materials and using workspace as agreed to in the PCM. This section also contains the site preparation review included in the PCM and indicates what preparation has been performed. Include the following in this section:

- Requirements for an inventory and inspection of the LOM, as well as any excess materials
- Instructions for redlining drawings, distributing drawing information, and approving changes to the installation procedures
- Directions for using spares to complete the installation
- Verification/QC/test responsibilities
- POCs for problems with material-developer-provided resources
- After action reports for deficiencies or problems identified

Example 2 lists implementation team responsibilities.

Example 2. Implementation Team Responsibilities

2.0 Implementation Team Responsibilities

a. Inventory and inspect the LOM prior to installation commencement, and notify the USAISEC project engineer of any missing or damaged items.

b. Make minor changes to the installation instructions without prior engineer approval. Minor changes do not alter the specified floor plan or major equipment list, violate a mandatory standard, alter the intended operational capability or procedures, alter cost or schedule, or alter the required testing intent or end result.

c. Obtain approval of the USAISEC Project Lead prior to making major changes in the installation instructions, including changes in equipment layout or operation and deleting or using different equipment.

d. Upon completion of the installation, the Implementation Team Chief will modify three sets of drawings to show the site, facility, building, or room as-built condition covered by this EIP. Denote modifications using red pencil for additions, yellow for deletions, and blue for notes to the draftsperson. One set of drawings will remain onsite with the O&M Command, and submit the remaining sets to the USAISEC Project Lead.

e. Inventory the excess materials after installation, and report the results to the USAISEC Project Lead. The inventory will include the part/model number, serial number, description, condition, and quantity of each item. The O&M Command will send an e-mail to the material fielder requesting disposition instructions and a fund citation if required.

f. Contact the USAISEC Project Lead for instructions if it appears a deficiency exists in the installation materials or equipment. Report the use of spare equipment to complete an installation to the USAISEC Project Lead.

g. Provide an after action report to the USAISEC Project Lead upon completion of the installation, identifying the problems occurring during the installation and the actions taken to resolve them.

h. Upon completion of the project, all trash associated with the installation shall be gathered in one area for O&M removal. Leave the premises in a condition acceptable to the Facility Chief.

2.3 POCs

This section lists POC information the implementation team may need for engineering, material development, project management, etc., as required. Example 3 lists the POCs.

Example 3. POCs

3.0 POCs

POCs for this project are as follows:

a. USAISEC POC: Ms. Mary Smith, Team Leader

Address: Commander, USAISEC
ATTN: ELIE-ISE-xx
Building 53301 Arizona Street
Fort Huachuca, Arizona 85613-5300

Defense Switched Network (DSN): 879-3000

Commercial Phone: (520) 538-3000

Commercial Fax: (520) 538-3041

E-mail: mary.smith.civ@mail.mil

b. Product Director, Satellite Communications Systems (PD SCS) POC: Mr. Joe Smith

Address: PD SCS
9351 Hall Road
Building 1456 2nd Floor, 2NW4906
Fort Belvoir, Virginia 22060

DSN: 656-0001

Commercial Phone: (703) 806-0001

E-mail: joe.smith.civ@mail.mil

c. O&M Command (Naval Satellite Communications Facility [NAVSATCOMFAC] Lago Patria) POC: LT Mike Jones, Officer in Charge (OIC)

Address: NAVSATCOMFAC
Lago Patria, Italy
PSC 822 Box 1
FPO AE 09621

DSN: 314-626-3357/3489

Commercial Phone: 011-39-081-568-3357/3489

Commercial Fax: 011-39-081-568-3504

DSN Fax: 314-626-350

E-mail: mike.jonesnotreal@eu.navy.mil

2.4 Appendices

The fourth section indentifies and describes the appendices included for the project. Example 4 lists the appendices.

Example 4. Appendices

4.0 Appendices

The following appendices complete this document:

- Appendix A. Installation Instructions
- Appendix B. Installation Drawings
- Appendix C. LOM
- Appendix D. Validation/Test Information

2.5 Installation Instructions (Appendix A)

Refer to drawings as much as possible when providing installation instructions in Appendix A of an EIP. Begin this appendix with general instructions for the project OSE, Team Chief, and implementation team. Include subjects, such as safety, site cleanliness, building access, work hours, and individual duties. All individual steps/instructions should be clear and concise. Most steps or instructions should refer to a drawing, publication, or some other documentation demonstrating how to perform the step. A picture is worth a thousand words, so use drawings as pictures.

Write clear instructions, so the implementation team using the EIP can install the equipment. Instructions should be clear enough for the implementation team to follow with minimal supervision.

Provide the installation instructions in modules as follows:

a. Equipment Layout – Address this first, as it is important to ensure the layout of the equipment is correct prior to the installation of any equipment. In large facilities installing large amounts of equipment, the Team Chief and OSE will ensure the engineer responsible for the project complies with the equipment layout.

b. Grounding Infrastructure – This instruction provides a drawing reference for and discusses the interior grounding installation that must take place prior to infrastructure installation and/or connectivity to the interior grounding installation from the existing cabinet/bay and or equipment installation.

c. Mechanical Infrastructure – This includes slotted channels, auxiliary framing, ducts/trays (for Alternating Current [AC], Direct Current [DC], power, signal, and fiber optics), equipment cabinets, floor tile cutouts, and all grounding required to support the mechanical infrastructure. In some instances, this also includes Heating, Ventilation, and Cooling (HVAC) equipment.

d. Power Infrastructure – This includes all AC power to the cabinets/equipment with panel schedules. This also includes the DC power from the main distribution panels to the sub-distribution panels and equipment.

e. Outside Plant (OSP) – This includes all signal and power cables installed outside the facility, whether in conduit or another conveyance. In many cases, this work may begin earlier, and a separate team may perform the work in parallel with the work.

f. Equipment Rack Elevations – This includes all rack and/or wall-mounted equipment, as well as grounding connections for the equipment. This also includes desk-mounted equipment, such as computer workstations.

g. Power Cabling – This includes power cabling to all equipment from AC distribution panels to power outlets/strips in the cabinets or individual pieces of equipment. This also includes all power cabling from the DC rectifier to the sub-distribution panels connecting to DC-powered equipment. If the equipment has an “A” and “B” power feed, ensure separate “A” and “B” sources are used. Also, ensure there are two separate power strips for the two AC feeds and two different circuit breakers on different sub-distribution panels for DC power. Label all sources with equipment information, and label all equipment with source information.

h. Signal Cabling – This includes all signal cabling (copper) to cabinet and wall-mounted equipment. Ensure all cables have separate “to” and “from” labels at each end of the cable.

i. Fiber Optic Cabling – This includes all fiber optic cabling to cabinet and wall-mounted equipment. Ensure all cables have separate “to” and “from” labels at each end of the cable.

2.6 Installation Drawings (Appendix B)

Appendix B of an EIP will include drawings on A- or B-size sheets for reference. Whenever possible, develop these drawings on A- or B-size sheets, since this improves the readability of the drawings when printing the document. The implementation team will receive three sets of drawings for use during the installation. When available, use standard drawings as much as possible. If project-specific drawings are required, include installation instructions on the drawings.

All drawings will comply with the most recent drafting SOP applicable to the project. The drawings in an EIP are for the installer; therefore, they should detail how to install the products. Indicate equipment installation details in written notes and in the specific details of each drawing. All drawings will have a list of parts with find numbers. Consider developing standard drawings for equipment, configurations, and infrastructure that could serve repeatedly in a project or program.

The development of as-built (or redlined) drawings takes place during the installation using the following methods of denotation:

a. Red ink will indicate changes.

b. Yellow highlight will indicate deletions.

c. Blue ink will indicate notes to the draftsperson.

Eventually, the as-built drawings will assist in the development of the plant-in-place drawings for the site as a record of what was installed and how it was installed.

The following list indicates the general order of drawings in an EIP; however, not all drawings are necessary for each EIP. The drawing package starts with an index. The index should include columns for a drawing number, revision number, title, and remarks. The following provides additional information on the typical drawings included in an EIP:

a. Drawing Index “DI” – This is the first sheet or drawing in each set. This is a complete list of drawings for a project, site, or facility. The drawing index is usually prepared on a B-size sheet or on several sheets, and it lists all of the drawings in the set by drawing number and title. This index shows the number of sheets for each drawing if more than one sheet is used, and it displays the revision of each drawing. Usually, only one drawing index is prepared for a site or facility; however, when the site is part of several major communications systems, more than one may be required. If a particular drawing is applicable to more than one system because of positioning, interconnects, etc., it will be listed on each of the respective drawing indexes.

b. System Diagram “SY” – This drawing shows the interrelationships and connectivity of the component elements of a system. This diagram uses labeled circles or rectangles to represent system elements and uses lines to represent connections. This diagram also uses symbols to portray special characteristics of system components (i.e., repeaters, passive reflectors, interfaces with other systems, etc.).

c. Base Layout Drawing “BA” – This is a to-scale drawing that provides the base boundaries, including the placement of properties, roads, streets, railroads, waterways, runways, and survey markers, as well as the location of the different Communications-Electronics (C-E) facilities. This drawing inserts a vicinity map to illustrate the physical relationship of the installation to the surrounding area. In addition, the base layout drawing proportions the vicinity map to size with the to-scale drawing. The installation layout drawing may contain a grid overlay to serve as the key sheet for OSP drawings. The grid should divide the base into rectangles that are 1,600 feet high and 2,400 feet wide. If it is necessary to show the details of a congested area on subsequent OSP drawings, the grid should use a larger scale, such as 800 feet by 1,200 feet. Grid lines should not cut primary buildings or C-E facilities. If this is not possible, do not divide these salient features into more than two parts. Details of each grid section will be contained on large-scale OSP drawings.

d. Site Plan "SP" – This drawing shows the overall site plan of the C-E facility and displays a plan view of the installation, to include the location and dimensions of major components and the location of survey markers, antennas, and antenna azimuths. A site plan may show manholes, ducts, underground and buried cables, aerial cables, and other OSP facilities when not provided on base grid or cable diagram drawings.

e. Architectural and Engineering Drawing "AE" – These drawings show the design and engineering required for the manufacture of specific items, such as structures, buildings, open areas, or equipment. Architectural engineering drawings encompass construction requirements, manufacturing specifications, material requirements, style, character, and aesthetic values. The information is definitive, precise, and to scale. The U.S. Army Corps of Engineers (USACE) can often provide this information for new buildings.

f. Building Layout Drawing "BL" – This is the key drawing for the family of drawings pertaining to a specific building. A building layout drawing is required for each building containing C-E facilities or portions thereof. This drawing must identify the room number and/or operational use of each room or area. The drawing notes state the type of building construction, floor composition, walls, and ceiling, as well as the floor capacity, ceiling heights, area floor space, and miscellaneous information. The plan view for this drawing is normally a single-line building outline showing all interior walls and partitions, and the plan view may show power ducts. The building layout drawing should be a to-scale drawing using as many sheets and match lines as necessary. Usually, this drawing requires one sheet for each floor. In this drawing, single lines represent inside walls, and breaks in the lines represent internal and external access openings of the building. Furthermore, symbols indicate electric, power, water, heat, and ventilation service entrances or internal source locations in a building layout drawing. This drawing identifies the major C-E facility and equipment locations, the supporting floor plans, and the location of the building as shown on the base layout map.

g. Cable Vault Drawing "CV" – This drawing shows the location, layout, racking, cabling, splicing, steps, access ports, and ceiling sleeves in a cable vault. This drawing also shows the relationship of the vault to the building and an elevation view showing length, width, headroom, and duct arrangement. In addition, a cable vault drawing shows the arrangement of cables on racks and the approximate location of valves, pressure plugs, sleeves, and splices, as well how the cables enter the vertical side of the Main Distribution Frame (MDF). This drawing must identify the cable or trunk number, cable count, and number of pairs. Prepare only one cable vault drawing per building with C-E functions.

h. Floor Plan "FP" – Use this drawing to support a building layout drawing. A floor plan provides detailed information about specific communication rooms. A floor plan is drawn with the maximum space available at the top of the sheet to leave room for an equipment legend and description. The drawing title must contain the numerical identification of the building, room number, and area, as well as the base and/or site name and address. A floor plan must contain a plan view of a room or area as identified by the building layout drawing. A floor plan also must show wall thicknesses, door swing directions, and access openings, as well as the location of all partitions, columns, and dimensions, to establish special limits in all directions. A floor plan identifies interior dimensions, numerical designations, and/or operational uses of all rooms or areas. Note any element of restriction to the installation, maintenance, or operation of C-E equipment on this drawing. Identify all C-E equipment installed or programmed onsite, and ensure the equipment is shown to scale and identified in the drawing legend. Heavier lines should indicate equipment fronts in a floor plan.

i. Rack Elevation Drawing "RE" – This is a dimensional, elevation view of equipment racks, consoles, and cabinets showing the equipment arrangement and blank panels. In this drawing, the item number must identify equipment when an equipment legend is used. A rack elevation drawing must show the shapes and sizes of compartments, assignments of space, and general mechanical and electrical details of component mounting. Note the building, room, equipment row, and operational use of the equipment on this drawing. Prepare one rack elevation drawing for each facility.

j. Wall Elevation Drawing "WE" – This drawing depicts all equipment. This includes furniture and accessories mounted on transportable van and shelter walls or on building walls.

k. Distribution Frame Drawing "DF" – This drawing shows the placement, layout, arrangement, and numbering of each MDF, Intermediate Distribution Frame (IDF), and Combined Distribution Frame (CDF). This drawing must identify the terminal blocks and cable run numbers of all cable runs to the frame, as well as terminal block designations. This drawing also includes the location and identification of protectors and terminal strips on a frame, whether these are in use or reserved for future use. This type of drawing should indicate set jacks and detail views presented when those terminal strips serve more than one circuit. Only one set of distribution frame drawings is normally prepared for each building. Each sheet of the drawing will show only one distribution frame.

l. Terminal Block Layout and Assignment Drawing "TB" – The installers and maintenance personnel use these drawings to identify each terminal block in a distribution frame (MDF, IDF, CDF, etc.), each terminal block's function, and where the other end of the cables terminate or originate. This data will include pictorial views of the blocks and frame, identification of the blocks, and pertinent information in narrative or tabular form. This drawing's data will be used in conjunction with the cable running list. The terminal block layout and assignment drawing should identify all spare blocks, terminals, wires, and cables.

m. Cable Rack Diagram "RD" – This drawing shows the construction and placement of cable racks, runways, raceways, trays, ducts, etc., used to support the interfacing cables. This diagram will use a plan view that shows the cable rack relative to equipment and building features. This diagram will also use an elevation view to supplement the plan view by using heavy lines to indicate the rack and light lines to indicate the equipment. A cable rack diagram should identify the type and size of components in the drawing (i.e., conduits, elbows, dropouts, and reducers) if engineering requirements demand such information.

n. Grounding System Drawing "GS" – These drawings show all the details necessary to depict all grounding systems associated with the facility, both inside and outside the building. Schematically, these drawings also show how and where cable shields, equipment cabinets, individual equipment, and other devices are grounded and bonded. Grounding system drawings show the entire communication facility's grounding system, including onsite power distribution, protective ground, lightning protective ground, signal ground, special Electromagnetic Pulse (EMP) protective ground, and an Earth Electrode Subsystem (EES). In addition, these drawings show all specialized hardware and techniques used to provide grounding, such as the use of chemicals, counterpoises, ground rods, wells, etc. Grounding system drawings should also display the EES's resistance-to-earth measurements and the methods of measurement, including the date of the last measurement.

o. Power Diagram "PD" – These drawings are specific in nature to identify power and the corresponding circuits. These diagrams are one-line diagrams that describe the AC and DC power systems, emergency and standby power, and safety/utility grounds (but not signal grounds), including special grounds installed for lightning and EMP protection. Power diagrams show all circuits, breakers, conduits, and other related hardware available for use in the power system. These diagrams also show power distribution to the equipment and cabinet(s), including the circuit assignments and power capacities of all outlets. In addition, power diagrams show unique features, such as uninterruptible power supply systems, automatic switchover systems, and monitor and control circuits. These diagrams should clearly identify any provisions required to expand the power system. Furthermore, power diagrams should include any tabular and narrative data necessary to describe the entire power system clearly.

p. Unistrut Plan "UP" – This drawing will provide a plan view of the unistrut installed in the technical equipment area. A unistrut plan is a superimposed drawing on the technical area rack layout. This plan provides a detailed view depicting the unistrut cross-section and model number (i.e., P-1000).

q. Installation Drawing "IN" – These drawings show how to mount, install, or place individual equipment at the facility. In addition, installation drawings provide details concerning the specific locations and actual securing of items. These drawings also provide supplemental information that is not available in the standard installation practice documents/drawings or manufacturer's documents.

r. Interconnect Diagram "ID" – This block diagram shows the electrical connections of a system, as well as its components, parts, and devices. This diagram also shows the actual interconnection of equipment when it differs from the functional interconnection shown on the block and level diagrams. This detailed drawing will assist in the development of the cable running list. The more accurate and detailed the information is on an interconnect diagram, the easier it will be to develop the cable running list. An interconnect diagram helps to ensure the cable running list is not missing any cables or interconnections between equipment.

s. Cable Running List "CL" – These lists assist in identifying, in tabular form, the origin and termination of all cables used in the facility (i.e., signal, power, protective ground, signal ground, radio frequency cables and waveguides, control, and intercom). Symbols and narrative data supplement cable running lists when necessary. The information shown in this list includes cable types, cable and wire sizes, cable functions, circuit assignments, identification coding, unused and spare wires and cables, and any special characteristics of the cables and wires. Use cable running lists in conjunction with interconnect diagrams. The cable running list will refer to standards and/or installation details that will tell the installer where to find a diagram for the cable assembly. This reference will include the drawing number, sheet number, and/or detail and table as required.

t. Circuit and Channel Assignment Drawing "CA" – This tabular drawing designates circuits and/or channel assignments to communications equipment that process more than one channel (i.e., patch panels, consoles controlling several radios, etc.). A diagram of the face of the equipment may assist in telling the installer what information to place on the label and where to place that label on the equipment being installed.

u. Cross-connect Diagram "XC" – This drawing depicts all permanent and semi-permanent wiring cross-connections between equipment on the distribution frame. This diagram includes the identification and location of equipment racks that interconnect.

v. Miscellaneous Detail Drawing "MS" – Use this drawing when requiring specific information that cannot be categorized in the detail codes of any other drawing listed in this document.

w. Standard Drawing "SD" – This is a common and repeatedly used drawing showing the preferred configurations of installation items. This drawing contains information used by installation personnel to install specific equipment or systems.

x. Audio-Visual Drawing "AV" – Use this drawing for details specific to audio-visual equipment and systems.

y. Outside Plant Drawing "OP" – This drawing shows the OSP, including buried cable, underground conduit, manholes, terminals, poles, and other features similar to OSP facilities. These features include cable numbers, number of pairs, pair counts, and terminal counts. Usually, this drawing is keyed to the grid system established by the base layout.

z. Manhole Detail Drawing "MH" – This drawing shows the information required to install underground cables in a specific manhole. This drawing should provide a description of the manhole, to include the number and dimensions, location of entry points, cable racks, and type of drainage. In addition, this drawing should contain the number, type, material, and inside diameter of conduits; the arrangements of cable; and the location of coil cases and other pertinent equipment for manhole use.

2.7 LOMs (Appendix C)

Appendix C of an EIP will include all LOMs. Organize the items in the LOMs alphabetically or by commonality. In addition, include a listing of any special installation tools required.

LOMs include all items needed to complete the installation. LOMs should include enough information for the Project Management Office or purchaser to acquire all materials with minimal interaction from the project lead. The following lists the fields that must be included in the LOMs:

a. Line Number: This information is for referencing the list while discussing the LOM over the phone or by e-mail.

b. Description: This field is one of the most important fields, and therefore, it should include as much information as possible to describe the part or piece of equipment.

c. Manufacturer/Company of Purchase: This information will allow the purchaser to know what company or distributor to contact to purchase the part.

d. Part and/or Model Number: This information will ensure the purchaser identifies the correct part/model.

e. Unit of Measurement (U/M): This information ensures the purchaser orders product using the correct measurement. Normally, cables are ordered specifying the number of feet or reels; however, hardware is ordered specifying a quantity per each unit.

f. Quantity (Qty): This information will ensure the purchaser orders enough of the product to perform the work.

Table 1 is an example of a LOM.

Table 1. Example LOM

BRAND NAME OR EQUIVALENT					
Item	**Description**	**Vendor**	**Catalog/Part Number**	**U/M**	**Qty**
1	RCM, High/Low-Density, Double Vertical Manager, 10" Wide, 7' High	Cooper B-Line	SB865610DFB	EA	2
2	Seismic-Rated Relay Rack, 19" Rack Mount, 84" High, Flat Black	Cooper B-Line	SB85219084FB	EA	2
3	RCM, High/Low-Density, Double Vertical Manager, 10" Wide, 7' High	Cooper B-Line	SB865610DFB	EA	1
4	Network Equipment Rack Junction Kit	Cooper B-Line	SB58704YZ	EA	2
5	Filler Panel, Black, 2 RUs (3.50")	Black Box	RMTB02	EA	4
6	RCM+ Horizontal Cable Manager, 19" Rack Mount, 2 RMU, Flat Black	Cooper B-Line	SB87019S2FB	EA	8
7	CAT 6, Feed-Through Patch Panel, 48-Port	Black Box	JPM816A	EA	12
8	Drawer, Storage, 2 RUs	Black Box	RMMT17	EA	2
9	Drawer, Storage, 3 RUs	Black Box	RMMT18	EA	2
10	Rack PDU, Basic, 1 RU, 20 A, 102, (10) 5-20; L5-20P	APC	AP9564	EA	2
11	CAT 6, Shielded Patch Cable, 26 AWG, Stranded, PVC, Gray, TBD ft. Long	Black Box	EVNSL0272GY-xx	EA	TBD
12	CAT 6, Shielded Patch Cable, 26 AWG, Stranded, PVC, Gray, TBD ft. Long	Black Box	EVNSL0272GY-xx	EA	TBD

A=Ampere; AWG=American Wire Gauge; CAT=Category; EA=Each; ft.=Foot; PDU=Power Distribution Unit; PVC=Polyvinyl Chloride; RCM=Rack Cable Management; RMU=Rack Mount Unit; RU=Rack Unit; TBD=To Be Determined

2.8 Validation/Test Procedures (Appendix D)

Ensure the physical installation of the equipment meets USAISEC standards by using the *Installation Workmanship Standards* document and filling in the QC inspection checklist (or site-specific checklist).

Include a test plan describing what testing the installers will perform and what acceptance testing the OSE will perform. This plan will include the test sequence, the required test equipment, and the standard procedures or references to standard procedures providing the details and minimum, allowable test results/readings (test specifications). Each test procedure shall include the test objectives, the equipment and materials required for testing (to include all cabling and interfaces), any special considerations as required, a list of references that support the test procedures, a step-by-step procedure telling the tester how to perform the test, and the test specifications that should be met. Use diagrams and sketches, as needed, to indicate the connectivity of the test equipment to the equipment tested.

3.0 CONCLUSION

Using the standards, references, and examples provided in this document, an engineer should be able to produce a quality product. Always ask peers, the Team Leader, the Critical Skill Expert, and the Group Leader questions when unsure.

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GLOSSARY. ACRONYMS AND ABBREVIATIONS

A	Ampere
AC	Alternating Current
AWG	American Wire Gauge
CAT	Category
CDF	Combined Distribution Frame
C-E	Communications-Electronics
DC	Direct Current
DISN	Defense Information Systems Network
DSN	Defense Switched Network
EA	Each
EES	Earth Electrode Subsystem
EIP	Engineering Implementation Plan
EMP	Electromagnetic Pulse
ETC	Earth Terminal Complex
ft.	Foot
HVAC	Heating, Ventilation, and Cooling
ICF	Interconnect Facility
IDF	Intermediate Distribution Frame
LOM	List of Materials
MDF	Main Distribution Frame
NAVSATCOMFAC	Naval Satellite Communications Facility
OIC	Officer in Charge
OSE	Onsite Engineer
OSP	Outside Plant
PCM	Project Concurrence Memorandum
PD SCS	Product Director, Satellite Communications Systems
PDU	Power Distribution Unit
POC	Point of Contact
PVC	Polyvinyl Chloride
QC	Quality Control
Qty	Quantity

RCM	Rack Cable Management
RMU	Rack Mount Unit
RU	Rack Unit
SDP	System Design Plan
SOP	Standing Operating Procedure
TBD	To Be Determined
TCF	Technical Control Facility
U/M	Unit of Measurement
USACE	U.S. Army Corps of Engineers
USAISEC	U.S. Army Information Systems Engineering Command